Study Report: OpenAI API Documentation - Generative AI Engineer

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Family:	Generative AI Engineer
Level:	Beginner
Chapters:	7
Source:	Reference: OpenAI API Documentation - OpenAI

Official sources and free libraries

- https://platform.openai.com/docs/

Summary

Study report for *OpenAI API Documentation* by OpenAI (Beginner level) mapped to the Generative AI Engineer role. Chat completions, embeddings, function calling, and best practices.

Chapters

Track: Book-Intro

Chapter 1: About OpenAI API Documentation [Beginner]

Chapter focus: About OpenAI API Documentation *(Level: Beginner)**

This study report summarizes *OpenAI API Documentation* by OpenAI for the Generative AI Engineer role. The resource is rated Beginner level. Chat completions, embeddings, function calling, and best practices. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# OpenAI API Documentation
# Author: OpenAI
# Role: Generative AI Engineer
# Level: Beginner
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is
A structured study report based on OpenAI API Documentation.

When to use it
When learning Generative AI Engineer skills at Beginner level.

Where to use it
Generative AI Engineer training paths and certification prep.

Real use example
A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Beginner]

Chapter focus: Why This Book Matters for Your Role *(Level: Beginner)**

Generative AI Engineer professionals use ideas from OpenAI API Documentation to solve real workplace problems. Chat completions, embeddings, function calling, and best practices. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

```
Role: Generative AI Engineer
Book focus: Chat completions, embeddings, function calling, and best practices.
Recommended level: Beginner
```

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is
Role-aligned learning connects theory to job tasks.

When to use it
During career planning and syllabus design.

Where to use it
Generative AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example
A teacher assigns this book report before a beginner skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Beginner]

Chapter focus: Key Topics Covered *(Level: Beginner)**

The main topics in OpenAI API Documentation include practical concepts described as: Chat completions, embeddings, function calling, and best practices. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Generative AI Engineer jobs today.

Code Reference:

```
- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept
```

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is
Topic maps turn a book into actionable learning objectives.

When to use it
Before reading and while building chapter summaries.

Where to use it
Self-study, flipped classroom, and revision.

Real use example
A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Generative AI Engineer Role Overview [Beginner]

Chapter focus: Generative AI Engineer Role Overview *(Level: Beginner)**

GenAI engineers build LLM-powered products: copilots, search, summarization, and codegen assistants. Skills span prompting, retrieval, evaluation, and safety - not just calling an API. 2025-2026 teams ship RAG before fine-tuning unless domain gap is proven.

Code Reference:

```
response = client.chat.completions.create(
  model='gpt-4o-mini',
  messages=[{'role':'system','content':'You are a helpful tutor.'},
            {'role':'user','content':'Explain SQL JOINs briefly.'}]
)
```

What it shows: System + user messages frame behavior - cheapest path to useful apps.

Topic guide

What it actually is
A GenAI engineer integrates LLMs into reliable user-facing applications.

When to use it
When unstructured knowledge or language tasks dominate the product.

Where to use it
Support bots, doc search, content generation, and code assistants.

Real use example
CodeQuest study copilot answers from official docs only - you own retrieval quality.

Chapter 5: Applied: LLM Fundamentals: Tokens and Context [Beginner]

Chapter focus: LLM Fundamentals: Tokens and Context *(Level: Beginner)**

Models read text as tokens (~4 chars English). Context window limits total input+output (128k+ on flagship 2025 models). Long prompts cost money and latency - summarize or retrieve instead of stuffing context.

Code Reference:

```
import tiktoken
enc = tiktoken.encoding_for_model('gpt-4o')
print(len(enc.encode('Hello, how many tokens am I?')))
```

What it shows: tiktoken estimates billable tokens before sending requests.

Topic guide

What it actually is
Tokenization and context limits shape every LLM application design.

When to use it
Sizing prompts, choosing models, and estimating cost.

Where to use it
All Chat Completions and embedding workloads.

Real use example
A 200-page PDF won't fit - chunking + RAG is mandatory not optional.

Chapter 6: Applied: Prompt Engineering Patterns [Beginner]

Chapter focus: Prompt Engineering Patterns *(Level: Beginner)**

Patterns: zero-shot, few-shot, chain-of-thought, JSON mode, structured outputs (2024+). Be explicit about format, length, and refusal conditions. Version prompts in Git.

Code Reference:

```
SYSTEM = """Answer using only provided context. If unsure, say I don't know.
Respond as JSON: {"answer": str, "citations": [int]}"""
```

What it shows: JSON schema reduces parsing bugs vs free-form answers.

Topic guide

What it actually is
Prompt engineering steers model behavior without weight updates.

When to use it
Every LLM feature before investing in fine-tuning.

Where to use it
Extraction, classification, tutoring, and support macros.

Real use example
Few-shot examples fix date format inconsistencies in invoice parsing.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Beginner]

Chapter focus: Study Plan and Practice *(Level: Beginner)**

Finish OpenAI API Documentation with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Generative AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is
Structured study plans improve retention and portfolio outcomes.

When to use it
After finishing the key topics chapters.

Where to use it
CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a beginner project screenshot.